

Getting Stuck Is Information

This year you will encounter questions you know how to solve immediately, questions you partially remember, questions you don't remember how to solve, questions you misunderstand, and questions you know how to solve but still get wrong.

These are different situations. They call for different responses.

The goal of this week is to move from “I got it wrong” toward “I know why I got it wrong, and I know what I can try next.”

The Five Questions

Use this routine whenever you're working on a problem, in math or anywhere else.

STOP

What information do I have? Identify the numbers, facts, definitions, labels, diagrams, or instructions in front of you. Don't calculate yet. First figure out what's actually available.

FIND

What am I trying to determine? Finish the sentence “I need to find ____.” This keeps you from calculating before you understand the goal.

CONNECT

What skill or information could help? Addition, a formula, information from a graph, a vocabulary definition. You don't need the whole solution yet, just something that might move you forward.

TRY

What strategy can I try? Write an equation, draw a diagram, estimate, eliminate impossible choices, reread the question, break it into smaller steps. The important word is try. A strategy doesn't need to be guaranteed to work before you attempt it.

CHECK

Does my answer make sense? Is it too large or too small? Should it be positive or negative? Did you answer the question that was actually asked?

Walk Through a Problem

A student earns \$15 per hour and works 6 hours. How much does the student earn before taxes?

STOP: \$15 per hour, 6 hours.

FIND: The total amount earned.

CONNECT: A rate multiplied by the number of hours.

TRY: $15 \times 6 = 90$.

CHECK: The student earns \$15 each hour. Six hours should be more than \$75 and less than \$100. \$90 fits.

Answer: \$90

This routine works outside of math too. Suppose you have a project due Friday that needs a written response, a visual, and a short presentation, and it's currently Wednesday.

STOP: three required components, due Friday, two days remaining.

FIND: how to finish the project on time.

CONNECT: planning and prioritization.

TRY: break the project into smaller tasks and decide what to complete Wednesday and Thursday.

CHECK: does the plan leave enough time for all three parts?

Why Was My Answer Wrong?

An incorrect answer doesn't automatically mean "I don't know how to do this." There are five different reasons an answer can be wrong, and each one calls for a different next step.

Error Types at a Glance

Knowledge error

You don't currently know or remember the information or skill needed. Example: you don't remember how to convert a fraction to a decimal. Next step: review the concept, look at a worked example, or ask for an explanation.

Process error

You understand the concept but didn't correctly complete the steps, or completed them in the wrong order. Example: you use the correct formula but substitute values into the wrong places. Next step: compare your work with a correct example and identify where the process changed.

Execution error

You knew what to do and used the correct process, but made a small mistake while carrying it out. Example: $7 \times 8 = 54$ instead of 56. Next step: slow down, check your calculations, and compare your answer with an estimate.

Comprehension error

You misunderstood what the question, passage, graph, or directions were actually asking. Example: you calculate the total when the question asked for the difference. Next step: reread the question and identify exactly what's being asked.

Strategy error

You understood the information and the goal, but chose an approach that was inefficient or unlikely to work. Example: doing a long calculation by hand when estimating would eliminate most of the answer choices in seconds. Next step: try a different approach, estimate, or ask "is there an easier way?"

Errors can overlap. A problem might involve more than one issue. The goal isn't finding a perfect label — it's answering: **what should I do differently next time?**

What Should I Do Next?

If the problem was...	Then...
a knowledge issue	learn or review the missing concept
a process issue	compare your steps with a correct process
an execution issue	check your work more carefully
a comprehension issue	reread and restate the question
a strategy issue	try a different approach

Different problems require different responses. Doing "more practice" isn't always the best solution if you don't know what kind of practice you actually need.

What Is a Diagnostic?

This week you'll complete an ACT Math baseline. It's diagnostic, meaning its purpose is to find your starting point, not to produce a grade. Attempting unfamiliar questions still gives useful information, even if you get them wrong. Don't randomly click answers just to finish. A wrong answer you actually attempted tells your teacher more than a blank one.

Reading Your Results

Instead of just a score, your results will be described using three categories:

Current strengths

Skills you currently demonstrate consistently.

Developing skills

Skills you partially demonstrate or seem to remember.

Priority skills

Skills that need instruction or real review.

This language will come back throughout the quarter, so it's worth getting familiar with now.

Quick Reference

STOP — what do I have?

FIND — what am I solving for?

CONNECT — what could help?

TRY — what's a strategy?

CHECK — does it make sense?

Knowledge: did I know the information or skill I needed?

Process: did I correctly follow the steps?

Execution: did I make a small mistake while doing something I knew how to do?

Comprehension: did I understand the question correctly?

Strategy: did I choose an effective approach?

Keep this page. You'll use it all year.